Ref. Ares(2023)6548540 - 27/09/2023

Symbolic Logic Framework for Situational Awareness in Mixed Autonomy

Project funded by the European Innovation Council and SMEs Executive Agency

D10.1 Portfolio Activities Report No. 1

Work Package: WP10

Authors: Arabinda Ghosh, Sadegh Soudjani

Project Title: Symbolic Logic Framework for Situational Awareness in Mixed Autonomy

Project Acronym: SymAware

Grant Agreement number: 101070802

Project Start Date: 1st October 2022

Duration: 36 months

Project Coordinator: Max-Planck-Gesellschaft Zur Forderung Der Wissenschaften EV (MPI-SWS)

Project Members:

Max-Planck-Gesellschaft Zur Forderung Der Wissenschaften EV (MPI-SWS)

Technische Universiteit Eindhoven (TUE)

Kungliga Tekniska Hoegskolan (KTH)

Uppsala Universitet (UU)

Stichting Koninklijk Nederlands Lucht - En Ruimtevaartcentrum (NLR)

Siemens Industry Software Netherlands BV (SISW)

Document Identifier: D10.1
Issue: 3.0
Issue Date: 26.09.2023
Prepared by: MPI-SWS
Approved: Yes
Dissemination Status: Sensitive
History Chart:

Issue	Date (DD/MM/YYYY)	Updated Sections	Cause of Change	Implemented by
1.0	01.09.2023	All sections	New Document	Arabinda Ghosh
2.0	10.09.2023	All sections	Revised the text and modified the information of events	Sadegh Soudjani
3.0	26.09.2023	Chapter 2	Revised section 2.1.8	Arabinda Ghosh Ana Tanevska

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1. Introduction

1.1 Deliverable Description

The first deliverable on portfolio activities reports the portfolio management and performance analysis related to the SymAware project. This report provides a comprehensive overview of the activities, performance, and developments within the portfolio for the reporting period M1 to M12. The objective of the portfolio activities is to enhance the likelihood of achieving successful project completion and initiating the preliminary stages of market transition by fostering synergies and partnerships among the initiatives supported through the 2021 EIC Pathfinder challenge call *Awareness Inside*.

The portfolio activities funded under 2021 EIC Pathfinder challenge call *Awareness Inside* encourages:

- Collaborations to overcome *shared challenges* within the Portfolio.
- Participation in regular meetings with the project officer (PO) to provide updates on ongoing technical tasks and achieved project objectives.
- Participation in outreach events at Portfolio level to facilitate connection with other researchers, industrial stakeholders, end users and showcase the technological progress, generated by the project.

2. Portfolio Activities

2.1 Achieved Objectives

Project members from TUE, UU and MPI-SWS have participated in multiple portfolio activities during the reporting period of M1-M12. The details of the events are as follows.

2.1.1 Portfolio Kickoff Meeting

Sadegh Soudjani from MPI-SWS and Zengjie Zhang from TUE participated in the portfolio kick-off meeting on 7-8 December 2022 at Saint-Josse-ten-Noode Brussels, Belgium. The following points were discussed during the meeting:

- Sadegh Soudjani presented an outline and main objectives for the SymAware project, as well as, key challenges, that can set the basis for the development of portfolio activities.
- Discussions were held on the objectives of the portfolio, strategic planning, methodology for executing the strategy, and focused activities (e.g., scientific, exploitation, communication, etc.) to execute the strategy.
- The project officer highlighted the key points to support the work at the portfolio level, by establishing several working groups, 1. steering committee (project coordinators), 2. on portfolio activities (portfolio managers), 3. on innovation and exploitation (exploitation managers), 4. on communication (communication managers), 5. on ethics (ethics advisors)
- Further discussions were held on the External Advisory Board for the portfolio.

2.1.2 Collaborations with EMERGE Project

Will Hendrix and Hiltje Nawijn from TUE had an online meeting with Renan Picoreti from EMERGE project to discuss issues related to communications activities in the portfolio.

2.1.3 Collaborations with ASTOUND Project

Sadegh Soudjani from MPI-SWS and Hiltje Nawijn from TUE had an online meeting with Isabelle Gordge, project manager of the EIC Pathfinder Astound project on possible collaboration to navigate the path from research project to business development, company building, and successful product commercialization.

2.1.4 Yearly Consortium Meeting of SymAware

The members of the SymAware project attended the yearly consortium meeting on 20–21 June 2023 at Eindhoven University of Technology in Eindhoven, the Netherlands. A summary of the key points covered during the meeting are enumerated below:

- The consortium members began by providing updates on the current status of their respective work packages within the SymAware project. This included sharing information about completed milestones, ongoing research activities, and the challenges encountered.
- Several breakout discussions consisting of a small group of partners were arranged. These smaller group sessions aimed to create collaborative opportunities.
- A discussion on software architecture was held. The consortium members successfully reached a consensus and finalized the software architecture for SymAware.
- MPI-SWS and UU are collaborating on a workshop paper on the agreed architecture. The main motivation of writing this workshop paper is to receive feedback from the community and then write a full paper on an extended edition of the architecture with the involved researchers.
- MPI-SWS and NLR are collaborating to integrate the UAV demonstrator into the current architecture of the SymAware.
- TUE and SISW are collaborating to develop the autonomous driving demonstrator and integrate the demonstrator into the current architecture of the SymAware.
- UU and SISW are collaborating to integrate the human agent in the autonomous driving demonstrator.
- MPI-SWS, TUE, and KTH are collaborating on the development of the initial version of the proposed software in SymAware.
- MPI-SWS and KTH are organizing meetings to synchronize on the joint research activities. Other frequent scientific meetings between “two partners” are being organized by the work package leads with frequency adapted to the needs of the progress of the activities.
- A discussion on the licensing of the software was also held. The appropriate licence is chosen according the promises in the project proposal.
- Matthijs Smakman, the independent ethics advisor, was present at the meeting and emphasized the significance of aligning the project’s objectives with ethical considerations. A discussion on several ethical questionnaires was held as a part of ethical deliverables D9.2 and D9.4.
- An extensive discussion took place regarding the dissemination activities of the project. The participants focused on strategies to enhance the outreach efforts and attract the interest of various industrial partners towards the project’s development. The discussion aimed to identify effective methods for raising awareness about the project and creating opportunities for collaboration with key stakeholders in the industry.

- The consortium members reiterated the importance of restructuring the current website to effectively engage potential stakeholders. The dissemination team of the project is currently working on the updates.
- The consortium members conducted a discussion regarding the organization of a workshop that would incorporate tutorials and hands-on sessions using the software developed by the project. Several prominent conferences focused on robotics and control were identified as potential venues for hosting this workshop. Further meetings and preparations will be undertaken to ensure the successful coordination of these workshops.
- A discussion on restructuring and rescheduling the general consortium meeting was held. It was collectively decided that the frequency of these meetings would be adjusted, with the new schedule being set for every four weeks instead of the previous bi-weekly arrangement.

The summary of the meeting was also shared with the project officer.

2.1.5 Steering Committee of Portfolio

Sadegh Soudjani from MPI-SWS attended the steering committee meeting with portfolio members on 12 April 2023 which was held online. The following topics were discussed during the meeting:

- Management structure of the portfolio and the composition of working groups (WGs).
- Arranging regular meetings of the WGs with decisions about frequency of meetings, election per WG of a chair and secretariat (in charge of setting the meetings, sending invitations, and drafting minutes).
- Decide the participation of the portfolio projects in ECAI 2023 in Krakow.
- The preparation for the work on the Strategic Roadmap of the Portfolio.

2.1.6 Kick-off Meeting of the Awareness Inside Working Group on Communications

Will Hendrix and Hiltje Nawijn from TUE participated in the kick-off meeting of the Awareness Inside working group on Communications which was held online on 28 June 2023. The following objectives were covered during the meeting:

- Discussions were held on the frequency of the meeting, and selection of chair, co-chair, and Secretary for the WG.
- Planning of workshop in ECAI 2023 (Krakow).
- Promotional aspects of the event reading the project and the portfolio.
- Funding aspects of portfolio events.

2.1.7 Kick-off Meeting of the Awareness Inside Working Group on Ethics

Ana Tanevska from UU participated in the kick-off meeting of the Awareness Inside working group on Ethics, which was also held online on June 28th. The kick-off meeting was led by the portfolio project officers, Carlos Casorran and Cyril Krykwinski.

The following topics were covered during the meeting:

- Deciding on the frequency of the meetings.
- Selection of chair (Ophelia Deroy (EMERGE)), co-chair (Martin Andraud (SUSTAIN)), and secretary (Michele Farisco (CAVAA)) for the WG.
- Planning of workshop in Brussels, September 20th, organized by EMERGE.
- Discussion of potential deliverables of the WG.
- Funding aspects of portfolio events.

The portfolio officers spoke more about the portfolio Steering Committee, and their work on the portfolio Strategic Plan. Following the finalization of the Strategic Plan, the WG can have a clearer idea of the roadmap of activities planned.

The WG will focus on the sharing of ethics activities and good practices in each project, as well as discussing the common challenges when dealing with ethic requirements and contexts.

2.1.8 EIC Portfolio Ethics Meeting

Ana Tanevska from UU and Arabinda Ghosh from MPI-SWS participated in the EIC Portfolio Working Group on Ethics Meeting which was held on September 20, 2023 at Brussels, Belgium. The two presented an overview of the SymAware project, as well as the ethical issues targeted in it through the research in WP6.

The following activities took place during the meeting:

- The opening remarks were given by Roumen Borissov, the portfolio officer, who highlighted the significance of ethical considerations for the projects within the portfolio.
- In addition to the presentations of the CAVAA, EMERGE, and ASTOUND project, the members of the SymAware project also shared their current endeavors and technological developments related to "awareness" and the ethical dimensions of their project. Valuable feedback was received (specifically regarding the ongoing participatory design user studies on UU's side, such as suggestions for questionnaires and qualitative metrics to be collected and analysed), and the SymAware project members are keen on implementation of these aspects into their work.
- Three group discussions were held based on the following topics:
  - RISKS: *What are the ethical red lines when it comes to awareness/consciousness in artificial agents?*;
  - BENEFITS: *Does awareness offer some opportunities to improve on the ethical guidelines for AI?*;

  - IMMEDIATE FUTURE: *What are the activities/initiatives that the Ethics portfolio could host in the next 12 months?*

- The program featured two keynote presentations as well:
  - *Towards an Ethical Governance Framework for (self) aware systems*, by Alan Winfield (University of Bristol)
  - *AI Enhancement: Can human intelligence be cognitively and morally enhanced by artificial intelligence?*, by Katleen Gabriels (Maastricht University)

2.1.9 Workshop on Awareness Inside: Open Meeting of the EIC Pathfinder Challenge

Sadegh Soudjani from MPI-SWS and Lefteris Vlahakis from KTH will participate in a portfolio workshop titled "Awareness Inside: Open Meeting of the EIC Pathfinder Challenge" on 1 October 2023 in ECAI 2023 at Krakow, Poland. A presentation on the project activities will be provided to the participants of the workshop.

3. Conclusion

In conclusion, the members of the SymAware project actively participated in several ongoing Portfolio Activities during the first reporting period to foster collaboration and to enhance the likelihood of achieving successful project completion. In future, we will be committed to continuously refine strategies and approaches to ensure the portfolio's continued growth and success.